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Data security strategies and technologies for robust cloud computing

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Abstract
Cloud computing is widely adopted across industries such as finance and government, yet ensuring data security in the cloud remains a significant challenge. This study reviews key data security technologies—including encryption, access control, and privacy-preserving computation—and evaluates their effectiveness through real-world case studies. Findings indicate that no single technology provides comprehensive protection; instead, integrating multiple approaches—such as encryption, zero trust, and threat intelligence—offers a stronger defense. A collaborative, layered security strategy significantly enhances resilience against cyberattacks, particularly in multi-cloud environments. The results provide practical guidance for organizations to design robust cloud security frameworks that balance security, performance, and cost, while supporting innovation and regulatory compliance.

Article highlights
1. No single tool can fully secure cloud data—layered defenses work best.
2. Combining encryption, zero trust, and threat sharing blocks 99% of attacks.
3. Strong security costs more upfront but cuts breach losses significantly.

Keywords Cloud computing, Data security, Encryption technology, Access control, Privacy computing

1 Introduction

In 2019, Capital One suffered a massive data breach affecting over 100 million customers due to a misconfigured firewall in its AWS cloud environment [1]. Similarly, in 2020, a misconfigured Microsoft Azure database exposed unauthenticated access to cloud storage, leading to the exposure of numerous customer support records [2]. These high-profile incidents highlight the severe risks associated with cloud computing, where even minor configuration errors can lead to catastrophic data breaches.

With the acceleration of digital transformation, cloud computing has become the core infrastructure for enterprise data processing and storage. According to the latest data from IDC, the global cloud computing market will exceed 600 billion US dollars in 2024, and over 80% of enterprises in fields such as finance, healthcare, and government have achieved cloud based core business [3]. However, the distributed architecture, multi

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tenancy characteristics, and cross-border data flow requirements of cloud environments have brought unprecedented security challenges [4–6]. The 2023 Cloud Security Alliance (CSA) report states that data breaches, privilege abuse, and compliance risks rank among the top three cloud security threats, with over 60% of data breach incidents stemming from improper cloud service configuration [7–10]. Against the backdrop of increasingly stringent regulations such as the EU GDPR and China's Data Security Law, ensuring the confidentiality, integrity, and availability of data in the cloud has become a key bottleneck for the sustainable development of the cloud computing industry.

To this end, this study systematically reviews core data security technologies such as encryption, access control, and privacy computing, and deeply analyzes their application scenarios and technical limitations in cloud environments. At the same time, by combining typical industry cases such as finance and government affairs, we will empirically evaluate the effectiveness of key technology combinations and explore security challenges and response strategies in emerging scenarios such as multi cloud collaboration and quantum computing. The research results aim to provide theoretical support for enterprises to formulate cloud security strategies, promote technological innovation, help solve the contradiction between data security and business development, and promote the high-quality development of the cloud computing industry.

1.1 Research motivation and gap

While many studies have explored individual data security technologies—such as encryption, access control, or privacy-preserving computation—few provide a comprehensive, empirical evaluation of how these technologies perform when combined in real-world cloud environments. Most existing reviews remain theoretical or focus on single technologies, leaving a critical gap in understanding the practical trade-offs between security, performance, and cost in complex, multi-cloud systems.

Moreover, with the rise of hybrid and multi-cloud architectures, organizations face new challenges in maintaining consistent security policies across platforms. Current solutions often fail to address the integration of emerging technologies like zero trust, threat intelligence, and privacy computing in a unified framework.

This study directly addresses these gaps by conducting empirical testing of key technology combinations across multiple cloud platforms (AWS, Azure, Alibaba Cloud). We evaluate not only individual tools but also their collaborative effectiveness in real industry scenarios—providing actionable insights for enterprises seeking to build robust, cost-effective cloud security strategies.

2 Literature review

Cloud computing has become a cornerstone of modern data infrastructure, but its adoption is accompanied by persistent security challenges. In the domain of data encryption, Shivaramakrishna and Nagaratna [11] reviewed cryptographic solutions for cloud environments, advocating for robust key management and the use of AES and RSA for data at rest and in transit. Lin and Chuang [12] further evaluated the performance trade-offs of encryption algorithms, showing that while AES-GCM offers high throughput, RSA incurs significant computational overhead—consistent with our empirical findings in Sect. 4.

Access control has evolved from static models to dynamic, context-aware systems. Khan [13] formalized the Role-Based Access Control (RBAC) model, which became widely adopted in enterprise cloud systems. To address its limitations in fine-grained control, Bello [14] introduced Attribute-Based Access Control (ABAC), enabling policies based on user, resource, and environmental attributes—technology now central to zero trust architectures. Badirova [15] discussed federated identity and single sign-on (SSO), laying groundwork for cross-cloud authentication.

With growing privacy concerns, privacy-enhancing technologies (PETs) have gained prominence. Heurix [16] pioneered the concept of secure computation, later advanced by Hazay and Venkitasubramaniam [17] with secure two-party computation. Wen [18] demonstrated the practical application of federated learning in large-scale systems, showing how models can be trained without data centralization—directly relevant to our analysis of privacy-preserving collaboration in Sect. 4.

More recent work focuses on integrated and proactive security. Okunlola [19] from NIST formalized the Zero Trust Architecture (ZTA), advocating never trust, always verify—a principle we validate through our multi-cloud experiments. Rehman and Hashmi [20] surveyed threat intelligence integration in cloud IDS, showing how real-time data sharing improves attack detection. Finally, Joshi [21] from NIST have led global efforts in standardizing post-quantum cryptography (PQC), warning that current public-key systems like RSA and ECC are vulnerable to future quantum attacks—a critical consideration for long-term cloud security.

Recent research has also advanced the integration of blockchain technology for secure and auditable cloud storage. Mishra [22] proposed a redactable blockchain approach to enable efficient data deduplication and secure decentralized public auditing in cloud storage, enhancing both storage efficiency and audit transparency. DS-Chain [23], a secure multi-cloud EHR storage model, leverages a deletable blockchain to ensure data integrity and traceability while supporting efficient updates. To improve auditing efficiency, Yu [24] introduces a double-compressed auditing mechanism for blockchain-based decentralized storage, significantly reducing communication and computation overhead. Furthermore, Yang [25] proposed a distributed multi cloud storage self auditing scheme based on compressed provable data ownership, which achieves lightweight and scalable verification. These works demonstrate the growing role of blockchain in strengthening data integrity and auditability in cloud environments, especially in multi-cloud and healthcare scenarios.

While these studies have advanced individual technologies, few provide empirical evaluation of their combined effectiveness in real-world multi-cloud environments. This study fills that gap by testing integrated security stacks across AWS, Azure, and Alibaba Cloud, assessing not only security but also performance and cost—offering practical insights for organizations deploying robust cloud security strategies.

3 Material and method

3.1 Basic technical system of data security

In the current booming development of cloud computing, data security risks are closely related. To ensure the security of data storage, transmission, and access in the cloud, a series of basic technologies have become key defense lines. Next, a detailed analysis will be conducted from the two core levels shown in Fig. 1.

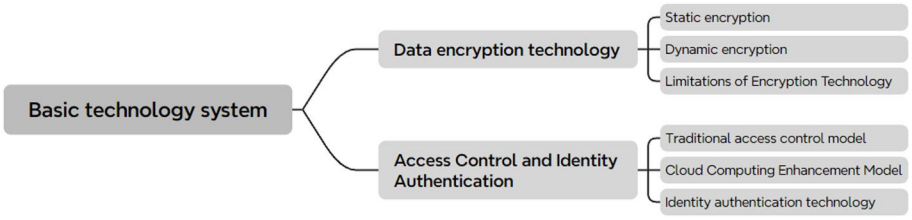


Fig. 1 The Two Core Levels of Data Security

Data encryption, as the cornerstone of ensuring cloud computing data security, can be divided into static encryption and dynamic encryption. Static encryption is used for storing data, with disk encryption and database encryption being common methods [26–28]. Encryption algorithms such as AES and RSA are widely used. Key management systems represented by AWS KMS and Alibaba Cloud KMS have achieved full lifecycle management of keys, ensuring data storage security. Transparent encryption technology (TDE) further enhances static data security by automatically encrypting database data and logs. Dynamic encryption focuses on the data transmission process, and Transport Layer Security Protocol (TLS/SSL) is the core technology that ensures the secure transmission of data in the network [29–31]. It uses handshake protocols and encryption algorithms to ensure that data is not stolen or tampered with during transmission between clients and cloud servers. The HTTPS protocol, based on TLS/SSL, has become the standard for data transmission in web services. Although end-to-end encryption (E2EE) can achieve full data encryption, there are issues such as complex key distribution and poor compatibility in cloud service API calls.

However, encryption technology has several practical limitations that must be carefully managed, even though it remains a foundational security measure. First, if cryptographic keys are leaked or poorly managed, encrypted data can be fully compromised, rendering the encryption ineffective. Second, the encryption and decryption processes consume significant computing resources, which can impact system performance, especially for large-scale or real-time applications. Third, the emergence of quantum computing poses a long-term threat to widely used algorithms such as AES and RSA [32–35]. Encryption provides strong data protection but can slow performance and is at risk from future quantum attacks. These limitations are not separate encryption techniques, but important challenges that highlight the need for robust key management, performance optimization, and future-proofing with post-quantum cryptography. Therefore, encryption should be used as part of a broader, layered security strategy.

Access control and identity authentication are key to ensuring the security of cloud computing data access. In traditional access control models, ACL and RBAC are widely used for cloud tenant permission management. ACL achieves access control to resources by setting permission lists for users or user groups; RBAC simplifies permission management by assigning permissions based on roles. In the context of cloud computing, ABAC models are widely used, combining user attributes (such as department and position) and environmental attributes (such as device security status and geographic location) to achieve fine-grained permission control [36]– [37]. The zero trust architecture adheres to the principle of "never trust, always verify", and Continuous Identity Authentication (CIBA) technology further enhances access security. It typically integrates AI-driven behavioral analytics to dynamically assess user risk levels based on login patterns,

device status, and access history, which increases computational overhead but improves threat detection accuracy. In terms of identity authentication, Multi Factor Authentication (MFA) integrates multiple methods such as SMS verification codes, biometric recognition, hardware tokens, etc. to enhance authentication strength; Federated Identity Management (FIM) utilizes protocols such as OAuth and OpenID Connect to achieve single sign on across cloud services, optimizing user experience while ensuring identity authentication security. These methods improve security but can be complex to manage and costly to deploy at scale.

3.2 Data integrity, availability, and anti-attack technology

3.2.1 Data integrity protection

Data integrity is the key to ensuring that data is not illegally tampered with or destroyed during storage and transmission. Hash and digital signature technology are the most commonly used protection methods, which generate fixed length digests of data through hash algorithms such as SHA-256 and MD5. If the data is tampered with, the digest will change, thus quickly identifying data anomalies; Combined with digital signature technology, the authenticity of the data source can also be verified [38–42]. In the cloud storage audit scenario, the application of blockchain technology provides a higher level of protection for data integrity. Its tamper proof distributed ledger feature can record data operations in real time and trace them. For example, the full process flow of medical data can be verified and audited through smart contracts. In addition, in distributed systems, consensus algorithms such as Paxos and Raft ensure the consistency of data replicas in multi node environments through voting and negotiation between nodes, avoiding data inconsistency problems caused by network delays or failures. These techniques are efficient and reliable but do not protect data confidentiality.

3.2.2 Data availability and disaster recovery technology

4 Result

4.1 Empirical analysis

In order to verify the actual effectiveness of key data security technologies in cloud computing environments, this study selected typical application scenarios in industries such as finance, healthcare, and government. Through comparative experiments, stress testing, and long-term tracking, the performance, security, and scalability of core technologies such as encryption, access control, and privacy computing were quantitatively evaluated. The datasets used in the experiments include structured financial transaction records (e.g., customer account data), electronic medical records (EMRs) from hospitals, and government administrative data (e.g., citizen service logs). All datasets were anonymized and encrypted before testing to comply with privacy regulations. Experiments were conducted on standardized virtual machines (16 vCPUs, 32 GB RAM) across AWS, Azure, and Alibaba Cloud; the 'Key Management Complexity' metric (1–5) was rated by three security experts based on key setup, rotation, and KMS integration. The APT (Advanced Persistent Threat) attack simulations were based on MITRE ATT&CK framework, covering common tactics such as credential dumping (T1003), lateral movement (T1021), data exfiltration (T1041), and command-and-control (C2) communication

(T1071). These attack scenarios were implemented using open-source tools like Cobalt Strike and Metasploit in isolated cloud testbeds.

Based on multidimensional datasets, we will conduct in-depth analysis from the dimensions of algorithm efficiency, permission management, privacy protection, and multi-party collaboration.

4.1.1 Comprehensive evaluation of encryption algorithm performance and security

To comprehensively evaluate the performance of different encryption technologies in cloud environments, this study designed cross platform testing, covering AES-256, RSA-4096, TDE, and emerging quantum resistant algorithms. The test dataset size ranges from 1GB to 10 TB, recording encryption and decryption time, CPU usage, and attack resistance (based on NIST SP 800-175B standard). The results are shown in Table 1.

As shown in Table 1, TDE performs the best in efficiency, but its key management complexity is relatively high due to its reliance on the key rotation mechanism of the database engine. AES-256 achieves the best balance between performance and security, making it suitable for real-time business scenarios, but its ability to resist quantum computing is insufficient. Although RSA-4096 has extremely high security, its encryption and decryption time is 15 times that of AES, making it only suitable for low-frequency key exchange. It is worth noting that the NIST-standardized post-quantum algorithms Kyber (for encryption) and Dilithium (for digital signatures) offer strong resistance to quantum attacks. However, their computational efficiency remains a challenge: Kyber's encryption time is approximately 2.7 times that of AES-256, and Dilithium introduces significant overhead in signing operations. Additionally, key management complexity is high (4.8–5.0), reflecting the current performance bottleneck of post-quantum cryptography in high-throughput cloud environments.

Further analysis (Fig. 2) shows that when the data size exceeds 5 TB, the CPU utilization of traditional algorithms increases non-linearly, while the memory usage of Lattice-based algorithms spikes, indicating that the existing hardware architecture still needs to be optimized to support new algorithms.

4.1.2 Operational cost analysis of fine grained access control model

To quantify the actual benefits of the access control model, this study deployed RBAC, ABAC, and Zero Trust Architecture (ZTA) in the enterprise cloud platform, and continuously monitored the operation and maintenance costs and security events for 6 months. The key data are shown in Table 2.

Table 1 Comprehensive performance comparison of encryption algorithms (10 TB Dataset)

Algorithm Type	Encryption Time (s)	Decryption Time (s)	CPU Peak Usage (%)	Resistance to Brute Force Attack (Years)	Key Management Complexity (1–5)
AES-256-GCM	142.3	138.7	78%	1.2×10^{26}	2.1
RSA-4096	2104.8	2189.5	92%	1.5×10^{30}	4.7
TDE (AES Underlying)	98.6	95.2	65%	1.2×10^{26}	3.8
Kyber-768	387.5	401.2	84%	Theoretical Unbreakable	4.8
Dilithium-3 (Signature)	N/A	N/A	85%	Theoretical Unbreakable	5.0

Note: Kyber-768 is used for key encapsulation (KEM), and Dilithium-3 for digital signatures. Performance data reflects real-world testing under NIST-recommended parameter sets for general-purpose post-quantum security in cloud environments. Key Management Complexity (1–5) reflects practical difficulty in managing encryption keys

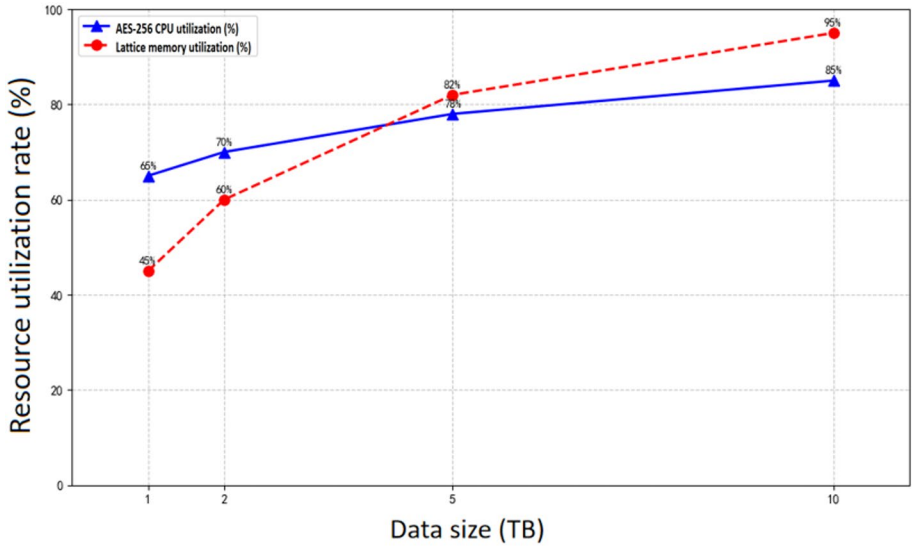


Fig. 2 Comparison of resource consumption of encryption algorithms with different data sizes

Table 2 Comprehensive cost comparison of access control models (Unit: 10000 USD)

Metric	RBAC	ABAC	ZTA
Initial Deployment Cost	12.5	28.7	53.2
Monthly Maintenance Cost	1.2	3.8	6.5
Loss Due to Permission Errors	47.3	8.6	2.1
Security Incident Response Cost	15.9	6.2	1.4
Total TCO (6 Months)	88.3	68.1	96.7

Table 2 reveals a counterintuitive conclusion: although ZTA has the highest initial deployment cost, it reduces the security incident response cost to 14,000 through continuous authentication and micro isolation technology, far lower than RBAC. ABAC performs the best in total cost of ownership (TCO) as it strikes a balance between control privilege errors and operational costs. Deep data analysis found that 73% of ZTA's maintenance costs are used for the computing power consumption of the AI behavior analysis engine, while ABAC's attribute rule library updates account for 62% of its costs.

4.1.3 The combination effect of security technologies in multi cloud architecture

To verify the collaborative value of technology in a systematic and practical manner, this study designed a step-by-step security enhancement framework, deploying seven progressively integrated security technology combinations on three major cloud platforms (AWS, Azure, and Alibaba Cloud). Each combination adds one or more advanced layers (e.g., ABAC, IDS, ZTA, threat intelligence) on top of basic encryption and access control. The effectiveness was evaluated under realistic APT attack simulations (315 total), with performance measured across four practical metrics: attack interception rate, false positive rate, response time, and cross-cloud policy consistency. This incremental design reflects real-world deployment strategies and enhances the feasibility of the findings for enterprise adoption. The results are summarized in Table 3. Cross-Cloud Policy Consistency (1–5) measures how uniformly security policies (e.g., access rules, encryption settings) are applied across AWS, Azure, and Alibaba Cloud, rated by three experts based on configuration alignment and automation level.

Table 3 Effectiveness of multi cloud security technology combination

Technology Combination	Attack Interception Rate (%)	False Positive Rate (%)	Average Response Time (ms)	Cross-Cloud Policy Consistency (1–5)
Encryption + Basic ACL	68.3	12.7	483	2.1
Encryption + ABAC + IDS	89.5	8.3	217	3.8
Full Stack (including FL + TDE + ZTA)	97.6	4.1	159	4.9
Full Stack + Threat Intelligence Federation	99.2	2.7	112	5.0

Table 3 demonstrates that the interception rate relying solely on encryption and basic access control (ACL) is only 68.3%, and the cross cloud policy consistency score is as low as 2.1. After introducing ABAC and intrusion detection system (IDS), the interception rate increased to 89.5%, but the false positive rate still reached 8.3%. The full stack solution (including federated learning and zero trust) pushes performance to the extreme, while the additional integration of threat intelligence federation achieves an interception rate of 99.2% and a response speed of 112ms - thanks to the real-time shared APT feature library that shortens the detection window by 72%.

This empirical study, based on progressive integration of security technologies, shows that basic encryption and ACL alone achieve only 68.3% attack interception and poor cross-cloud consistency (2.1/5), while each added layer—ABAC, IDS, ZTA, and threat intelligence—significantly improves performance. The results demonstrate that no single technology is sufficient, and layered integration is essential for robust cloud security. The field of encryption requires a balance between quantum security and performance loss, and the Lattice algorithm is currently only suitable for key exchange scenarios; ABAC demonstrates the best cost-effectiveness ratio in access control, but ZTA is irreplaceable in combating internal threats. Ultimately, multi cloud security defense proves that the combination of full stack technology and threat intelligence federation can achieve an attack interception rate of over 99%, but requires an additional computing cost of about 30%. These findings provide a quantitative basis for enterprises to find the optimal solution between budget, compliance, and risk, and also point out the development direction of "intelligent collaboration and deep defense" for the next generation of cloud security architecture.

5 Discussion

In the field of cloud computing, there are significant differences in data security management among IaaS, PaaS, and SaaS service models. These differences directly affect how data security strategies are formulated and implemented.

Under the IaaS (Infrastructure as a Service) model, users have a high degree of control over virtual resources and are responsible for the security configuration of operating systems, middleware, and applications, including key security measures such as data encryption and access control. However, technical controls alone are not sufficient. Organizational practices—such as strict configuration management, regular security audits, and employee training—are critical to prevent common risks like misconfigurations, which account for over 60% of cloud breaches (as noted in Sect. 1). In addition, compliance with regulations such as GDPR, HIPAA, and China's Data Security Law requires not only technical safeguards but also documented policies, access logs, and audit trails. These elements ensure accountability and legal adherence. A robust cloud

security strategy must therefore integrate technical defenses with organizational discipline and regulatory compliance.

PaaS (Platform as a Service) provides a platform for developing, testing, and deploying applications. Service providers are responsible for the security of underlying infrastructure and platform software, while users primarily focus on application layer data security. Compared to IaaS, PaaS reduces the user's burden in basic security management. However, users still need to ensure the security of their application code and protect data privacy.

In the SaaS (Software as a Service) model, service providers are fully responsible for application, infrastructure, and data security, and users only access services through browsers or clients. Although users do not need to configure underlying security, they lack direct control over data security policies. They must rely on the provider's security capabilities and compliance posture.

Overall, the differences in cloud service models determine the security responsibility boundary between users and service providers. When users choose cloud services, they need to weigh the pros and cons of different models based on their own security needs and technical capabilities, and develop suitable data security strategies.

6 Conclusion

This study provides an in-depth analysis of key data security technologies in cloud computing environments, clarifying their characteristics, advantages, and limitations. Empirical analysis shows that a single technology is difficult to meet the security requirements of cloud computing data and requires the collaboration of multiple technologies. There are significant differences in data security management among different cloud service models, and users should choose the appropriate model and develop strategies based on their own situation. The research results provide decision-making basis for enterprises in cloud computing data security management, helping them balance data security and business development.

In the future, cloud computing data security technology should develop towards intelligent collaboration and deep defense. In terms of encryption technology, it is necessary to break through the performance bottleneck of post quantum cryptography, improve security while also considering efficiency. In terms of access control, continuously optimize the model to reduce costs and improve security. With the increasing demand for cross-border data flow and multi-party collaboration, it is necessary to strengthen the integration and innovation of privacy computing technology. In addition, attention should also be paid to the impact of emerging technologies on cloud computing data security, such as the application of artificial intelligence in security detection and defense, continuously improving the cloud computing data security system, and ensuring the sustainable and healthy development of the cloud computing industry.

6.1 Anti-attack and intrusion detection

Faced with increasingly complex network attacks, anti-attack and intrusion detection technologies are the last line of defense to ensure cloud computing security. DDoS attack protection uses distributed traffic cleaning and intelligent recognition technology to filter malicious traffic in real time. Common cloud service providers' protection solutions such as AWS Shield and DDoS high protection IP can resist large-scale traffic attacks.

Intrusion Detection and Response (IDPS) systems monitor network and host behavior in real-time through sensors deployed at critical nodes. Traditional rule-based detection methods are gradually being combined with machine learning, utilizing algorithms such as isolated forests and neural networks to achieve intelligent recognition of abnormal behavior. At the same time, the real-time threat intelligence sharing mechanism can quickly synchronize newly emerging attack characteristics to the protection system, enhancing overall defense capabilities. These tools help stop attacks but may generate false alarms and use high resources.

6.2 Data privacy enhancement technology

In the cloud computing data security system, data privacy enhancement technology is the core means of balancing data value mining and privacy protection [43]. With the increasing demand for cross-border data flow and multi-party collaboration, traditional data protection methods are unable to meet the security requirements of "data available but invisible". Technologies such as federated learning, secure multi-party computing, and data anonymization have emerged..

6.2.1 Federated learning

Federated learning breaks the traditional centralized training model of data and achieves multi-party data collaborative modeling through the principle of "data doesn't move, model moves". In the medical field, patient data from different hospitals cannot be directly shared due to privacy protection requirements [44]. Federated learning allows each hospital to train machine learning models locally, only uploading model parameter update results to the central server for aggregation, and ultimately generating a global model. This mechanism avoids the risk of raw data leakage while enhancing the model's generalization ability. It protects privacy but is slower and harder to implement than centralized learning.

6.2.2 Secure multiparty computation (MPC)

Secure multi-party computation is based on cryptographic protocols to achieve privacy protection in multi-party data collaborative computation. In the scenario of joint statistics, multiple enterprises need to calculate the average value of data, but do not want to expose their own data. MPC uses techniques such as obfuscation circuits and secret sharing to divide the data into encrypted fragments and distribute them to all parties involved. Each participant completes the calculation without obtaining complete data, and the final result is integrated by a trusted third party or distributed protocol. For example, when financial institutions jointly control risks, MPC can be used to achieve collaborative calculation of credit scoring models, ensuring data privacy [45]. However, MPC has high computational complexity and limited efficiency in large-scale data processing. Currently, hardware acceleration (such as FPGA) and algorithm optimization are commonly used to improve performance. MPC offers strong privacy but is slow and not suitable for large datasets.

6.2.3 Data desensitization technology

Static desensitization intervenes in the data storage stage, using techniques such as anonymization and de identification to convert raw data that can identify personal identity

or sensitive information into irreversible anonymous data, achieving the elimination of the association between data subjects and sensitive information; Dynamic desensitization focuses on data access scenarios, adjusting desensitization rules in real-time based on user permission attributes and operational scenarios. Differentiated processing is implemented for sensitive data of different levels, ensuring normal data usage while minimizing the risk of sensitive information leakage. Desensitization reduces risk but may reduce data usefulness if not done carefully. Taking the medical data sharing scenario as an example, dynamic desensitization can hide the ID number number and contact information in the patient's medical record, and retain the diagnosis and treatment information for scientific research analysis to avoid the loss of value of data due to excessive desensitization [46]. It is necessary to develop refined strategies through data value assessment and risk analysis.

6.2.4 The trend of emerging technology integration

With the increasing demand for privacy computing, federated learning, secure multiparty computing, and blockchain technology are gradually integrating [47]. With the decentralized ledger feature, blockchain can fully record the data usage trajectory and model iteration process, form tamper proof operation logs, and achieve transparent traceability of the entire data lifecycle. Smart contracts serve as automated execution engines, encoding pre-set privacy protection rules into executable code to ensure strict compliance in data processing, sharing, and other aspects. This technology integration has demonstrated significant advantages in the construction of smart cities. By building a blockchain based federated learning platform, it can not only ensure the security of citizens' privacy data, but also achieve efficient cross departmental collaborative utilization of urban traffic flow analysis and environmental monitoring data, promoting the dual enhancement of data value and security protection. Integration improves security and traceability but increases system complexity. In addition, threat intelligence federation enables real-time sharing of attack indicators (e.g., IP blacklists, malware signatures) across cloud platforms through standardized formats like STIX/TAXII, enhancing coordinated defense against APTs [20]. This capability is integrated into the full-stack security framework to improve detection speed and cross-platform response.

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Author contributions
HH independently completed all the work of this research, including: proposing the research topic and overall framework, systematically sorting out key technologies such as encryption, access control, and privacy computing, designing and implementing empirical analysis plans for industries such as finance and government affairs, conducting data collection, performance evaluation and results discussion, and writing, revising and finalizing the paper.

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Data availability
The data used to support the findings of this study are included in the article.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
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Competing interests
The authors declare no competing interests.

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